

Le Nhu Dat

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SUMMARY

AI Engineer focused on LLM agents and RAG systems that automate knowledge-heavy work. 1+ year building RAG and Agent pipelines end to end: hybrid retrieval, reranking, answer verification, evaluation and observability.

PROJECTS

Vietnamese Legal QA – Production RAG

github.com/lndat18/production-legal-qa-rag

- Built an end-to-end RAG chatbot answering Vietnamese labor, insurance and tax-law questions with article-level citations over 6 legal documents (2,228 chunks), replacing manual legal text lookup
 - Preserved legal structure from ingestion to answer: converted `.docx` laws to structured Markdown and chunked by Article/Clause/Point with breadcrumbs, so every passage stays identifiable and citable
 - Designed a hybrid retrieval pipeline: dense + BM25 with Reciprocal Rank Fusion, HyDE, citation-aware candidate search, optional MMR, and a local Vietnamese GPU reranker
 - Added a fail-closed verification layer (deterministic citation and number checks + an independent LLM evidence judge, at most one repair) that refuses to answer when evidence is not supported
 - Engineered serving reliability: Redis answer/retrieval cache with single-flight, conversation follow-up handling, admission control, per-user rate limits, and throttling across multiple Groq API keys
 - Shipped an OpenAI-compatible FastAPI service with an OpenWebUI chat interface, deployed on a single host with Docker Compose and Cloudflare Tunnel, with automated backups
 - Instrumented every turn with Langfuse traces and Prometheus/Grafana metrics
 - Built a RAGAS evaluation pipeline: generated 203 synthetic cases, reviewed them down to a 157-question golden testset (142 single-hop, 15 multi-hop), and compared MMR on/off with checkpointed, resumable runs that handle API quota limits
 - Enforced code quality with pytest, Ruff, mypy and GitHub Actions CI, with a written spec for every pipeline stage
- LLM & Retrieval:** LangChain, Groq (gpt-oss-120b, gpt-oss-20b), Hugging Face, Pinecone, BM25, PyTorch
Backend & Deployment: Python, FastAPI, Redis, PostgreSQL, OpenWebUI, Docker, Cloudflare Tunnel
Observability: Langfuse, Prometheus, Grafana
Evaluation & CI/CD: RAGAS, pytest, Ruff, mypy, GitHub Actions

Agentic GraphRAG for Legal Q&A — In Progress

github.com/lndat18/production-legal-agentic-graph-rag

- Extending the RAG system above into a LangGraph agent that plans, retrieves, calls tools and verifies, to answer legal questions spanning multiple clauses and articles
 - Built a legal knowledge graph in Neo4j that models cross-references between articles, clauses and points, so retrieval can follow legal citations across documents
 - Designing three cooperating retrieval agents (retrieval planner, hybrid retriever, graph retriever) that decompose a request into subqueries and gather connected provisions
 - Designing a personal income tax agent: a deterministic calculator owns the arithmetic and the LLM extracts parameters, producing a per-bracket breakdown with legal citations
 - Designing document generation (tax forms as DOCX/PDF), effective-date checks for provision applicability, and scheduled law-change monitoring, with user confirmation before any external action
 - Planning agent evaluation with DeepEval and RAGAS: tool-call accuracy, task success rate, numeric error and cost
- Agent & Graph:** LangGraph, Neo4j
LLM & Retrieval: LangChain, Groq (gpt-oss-120b, gpt-oss-20b), Hugging Face, Pinecone, BM25, PyTorch
Backend & Deployment: Python, FastAPI, Redis, PostgreSQL, OpenWebUI, Docker, Cloudflare Tunnel
Observability: Langfuse, Prometheus, Grafana
Evaluation & CI/CD: RAGAS, pytest, Ruff, mypy, GitHub Actions
Planned: MCP, Jev (routing), DeepEval

EXPERIENCE

AI Engineer Intern

Ho Chi Minh City, Vietnam

TMA Solutions – software company

March 2026 – May 2026

- Developed a Sign Language Recognition solution for the hearing-impaired using CorrNet and MediaPipe keypoints
- Improved the preprocessing pipeline to handle variable-length sequences and missing keypoints, improving the score from 90% to 94% and reducing word error rate (WER) to 7% on the public How2Sign dataset
- Tuned model architecture and hyperparameters to further improve recognition performance

EDUCATION

TON DUC THANG UNIVERSITY

Ho Chi Minh City, Vietnam

Bachelor of Computer Science

2022 – 2026

GPA: 3.06/4.00 Classification: Good

CERTIFICATIONS

- **IBM RAG and Agentic AI** — IBM
- **IBM Generative AI Engineering Professional Certificate** — IBM
- **LLMOps (Large Language Model Operations)** — Duke University
- **MLOps (Machine Learning Operations)** — Duke University
- **Retrieval Augmented Generation (RAG)** — DeepLearning.AI
- **Natural Language Processing Specialization** — DeepLearning.AI
- **Deep Learning Specialization** — DeepLearning.AI
- **Machine Learning Specialization** — DeepLearning.AI
- **Introduction to Git and GitHub** — Google

ADDITIONAL INFORMATION

Programming Languages: Python, C, Java

Language Skills: IELTS 5.5 (CEFR B2)